

Software Requirements Specification (SRS)

Artisan Store Web Platform

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Project: Artisan Store Web Platform

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Prepared By: Requirements Team

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines all functional, non-functional, and business requirements for the Artisan Store Web Platform. It serves as the contract between the business owner and the development team, detailing what the system must do and how it must perform.

1.2 Scope

The scope of this SRS covers:

- Customer-facing web application features
- Administrator dashboard features
- Database and data management requirements
- System performance and security requirements
- Integration requirements (if any)

1.3 Audience

- Development team members
- Project manager
- QA testers
- Business owner
- System architects

1.4 Document Organization

1. Overall Description
2. Functional Requirements

3. Non-Functional Requirements
 4. Data Requirements
 5. User Interface Requirements
 6. Security Requirements
 7. Performance Requirements
 8. Business Rules (Enforcement)
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2. Overall Description

2.1 Product Perspective

The Artisan Store Web Platform is a standalone, web-based e-commerce application. It is not integrated with existing systems but may connect to email services for notifications.

2.2 Product Functions

- Display product catalog with images and descriptions
- Allow customers to search and filter products
- Enable customers to create and manage orders
- Support product customization requests
- Provide order status tracking
- Manage customer accounts
- Provide administrative tools for business management
- Generate sales reports

2.3 User Classes and Characteristics

Customer Users

- **Technical Skill Level:** Low to Medium
- **Experience:** May not be tech-savvy
- **Needs:** Easy product browsing, clear ordering process, order tracking
- **Access:** Web browser (desktop, tablet, mobile)
- **Frequency:** Occasional (when shopping)

Administrator Users

- **Technical Skill Level:** Medium to High
- **Experience:** Business owner or staff

- **Needs:** Efficient product management, order processing, reporting
- **Access:** Web browser (desktop)
- **Frequency:** Daily

2.4 Operating Environment

Supported Browsers:

- Google Chrome (latest 2 versions)
- Mozilla Firefox (latest 2 versions)
- Safari (latest 2 versions)
- Microsoft Edge (latest 2 versions)

Supported Devices:

- Desktop computers (1024x768 minimum resolution)
- Tablets (iPad, Android tablets)
- Mobile phones (iPhone, Android phones)
- Screen sizes from 320px (mobile) to 1920px (desktop)

Server Environment:

- Web server: Node.js/Express or Python/Django
 - Database server: PostgreSQL or MySQL
 - Hosting: Cloud provider (AWS, Azure, DigitalOcean)
 - Network: Internet connectivity required
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3. Functional Requirements

3.1 User Authentication & Account Management

FR-1: User Registration

- **Requirement:** The system shall allow new customers to create an account
- **User Story:** US-11 (Register)
- **Description:** New visitors must be able to register using email and password
- **Acceptance Criteria:**
 - Email must be validated as unique
 - Password must meet security requirements (min 8 characters)
 - Password confirmation must match original password

- Confirmation email must be sent to provided address
- Account is activated after email confirmation
- Error messages for invalid input are displayed

FR-2: User Login

- **Requirement:** The system shall authenticate users with email and password
- **Description:** Registered users must be able to log into their accounts
- **Acceptance Criteria:**
 - Email and password are validated against database
 - Incorrect credentials show appropriate error message
 - Session is created upon successful login
 - User is redirected to account dashboard
 - Login attempt logging for security

FR-3: Password Security

- **Requirement:** The system shall securely store and validate passwords
- **Description:** Passwords must be hashed and cannot be retrieved
- **Acceptance Criteria:**
 - Passwords are stored using bcrypt or similar algorithm
 - Password hashing is performed server-side
 - Password reset via email is available
 - Passwords must be changed every 90 days (optional requirement)

FR-4: User Profile Management

- **Requirement:** Customers shall be able to view and update their profile information
- **User Story:** US-11, US-13
- **Description:** User profile must show and allow editing of contact information
- **Acceptance Criteria:**
 - Profile displays: name, email, phone, address, province
 - All fields can be edited by customer
 - Changes are saved to database immediately
 - Email uniqueness is validated on update
 - Success/error messages are shown

3.2 Product Catalog Management

FR-5: Display Product Catalog

- **Requirement:** The system shall display available products organized by category
- **User Story:** US-01
- **Description:** Customers must see all active products grouped by their category
- **Acceptance Criteria:**
 - Products display with image, name, price, stock status
 - 3 categories are available: Wood, Ceramic, Amigurumis
 - Only products with "Active" status are shown
 - Products are paginated (10-20 per page)
 - Category selection is easy and intuitive

FR-6: Product Search and Filtering

- **Requirement:** The system shall allow customers to search and filter products
- **User Story:** US-02, US-03
- **Description:** Customers must be able to find products quickly using search and filters
- **Acceptance Criteria:**
 - Search by product name (case-insensitive)
 - Filter by category
 - Filter by price range
 - Filter by availability (in stock / out of stock)
 - Multiple filters can be combined
 - Search results show relevant matches first

FR-7: Product Detail View

- **Requirement:** The system shall display complete information about a specific product
- **User Story:** US-02
- **Description:** Customers must see detailed information about a product
- **Acceptance Criteria:**
 - Product name, description, price are displayed
 - Full product description (500+ characters)
 - Current stock quantity is shown
 - "Allows customization" flag is displayed
 - Rating/reviews (optional for future)

- Related products (optional for future)

FR-8: Product Photo Gallery

- **Requirement:** The system shall display multiple photos for each product
- **User Story:** US-03
- **Description:** Customers must be able to view high-quality images of products
- **Acceptance Criteria:**
 - At least 1 photo per product (required)
 - Multiple photos can be displayed
 - Photos display in gallery format
 - Zoom/enlarge feature available
 - Photos are optimized for web (< 500KB each)
 - Thumbnail navigation available

FR-9: Product Management (Admin)

- **Requirement:** Administrators shall be able to manage the product catalog
- **User Story:** US-14
- **Description:** Business owner must be able to add, edit, and delete products
- **Acceptance Criteria:**
 - Admin can create new product with all required fields
 - Admin can edit existing product information
 - Admin can soft-delete products (remove from catalog)
 - Admin can change product status (Active/Inactive)
 - Admin can upload product photos
 - Admin can mark if product "Allows customization"
 - Form validation shows all errors clearly

3.3 Order Management

FR-10: Create Order

- **Requirement:** The system shall allow customers to create and submit orders
- **User Story:** US-04
- **Description:** Customers must be able to add products to cart and submit order
- **Acceptance Criteria:**
 - Customers can select products and quantity
 - Minimum 1 item required per order

- Out-of-stock items cannot be added
- Subtotal is calculated correctly
- Shipping cost is calculated automatically
- Order total is displayed before confirmation
- Customer contact information is required and validated

FR-11: Calculate Shipping Cost

- **Requirement:** The system shall automatically calculate shipping costs based on province
- **User Story:** US-05
- **Description:** Shipping cost depends on whether customer is in same province as business
- **Acceptance Criteria:**
 - Same province: no additional charge
 - Different province: add configured cost
 - Cost is configurable by admin
 - Cost is displayed in order summary
 - Cost is itemized in order confirmation

FR-12: Generate Order Reference Number

- **Requirement:** The system shall generate unique reference number for each order
- **User Story:** US-06
- **Description:** Each order must have unique identifier for tracking and communication
- **Acceptance Criteria:**
 - Reference number is generated automatically upon order creation
 - Format: [Date]-[RandomString] or sequential number
 - Reference number is unique and never repeats
 - Customer receives reference number via email
 - Customer can use reference number to track order

FR-13: Order Confirmation

- **Requirement:** The system shall send order confirmation to customer
- **User Story:** US-06
- **Description:** Customer must receive confirmation of order submission
- **Acceptance Criteria:**

- Email is sent to customer with order details
- Email includes reference number
- Email includes all ordered items and quantities
- Email includes total price breakdown
- Email confirms expected delivery timeframe
- Email provides contact information for questions

FR-14: Track Order Status

- **Requirement:** Customers shall be able to check the status of their order
- **User Story:** US-07
- **Description:** Customers must be able to view current order status
- **Acceptance Criteria:**
 - Order can be found by reference number
 - Status displays: Confirmed, Paid, Processing, Ready, Delivered
 - Current status is clearly highlighted
 - Date/time of each status is shown
 - Only the customer can see their own orders

FR-15: View Order History

- **Requirement:** Registered customers shall be able to see all their previous orders
- **User Story:** US-12
- **Description:** Customer account must show complete order history
- **Acceptance Criteria:**
 - All customer's orders are listed
 - Orders show: date, reference number, status, total amount
 - Orders can be filtered by date range
 - Orders can be filtered by status
 - Clicking order shows full details
 - Order count is displayed

FR-16: Manage Orders (Admin)

- **Requirement:** Administrators shall be able to view and manage all orders
- **User Story:** US-15
- **Description:** Business owner must be able to see all orders and update status
- **Acceptance Criteria:**

- Admin can view all orders received
- Admin can filter orders by status, date, customer
- Admin can click order for full details
- Admin can change order status manually
- Admin can see customer contact information
- Admin can see all items in order
- Admin can view order notes/special requests

FR-17: Mark Order as Paid

- **Requirement:** System shall allow admin to confirm payment and start production
- **User Story:** US-15 (related: BR-09)
- **Description:** When payment is received, admin marks order as paid to begin production
- **Acceptance Criteria:**
 - Admin can change status to "Paid"
 - Only unpaid orders can be marked as paid
 - Status change is logged with timestamp
 - Customer is notified of status change
 - Order enters production queue after marking paid

3.4 Product Customization

FR-18: Request Product Customization

- **Requirement:** Customers shall be able to request customized versions of products
- **User Story:** US-09
- **Description:** Some products allow customers to request custom specifications
- **Acceptance Criteria:**
 - Only products marked "Allows customization" show customization option
 - Customization form captures: type, details, description
 - Customer can upload reference images (optional)
 - Order is marked as "Customized" in system
 - Submission triggers admin notification

FR-19: Manage Customization Requests (Admin)

- **Requirement:** Administrators shall review and approve customization requests
- **User Story:** US-10

- **Description:** Admin must approve custom orders and set final pricing
- **Acceptance Criteria:**
 - Admin sees list of pending customization requests
 - Admin can view full customization details
 - Admin can ask for more information (send email)
 - Admin can approve request
 - Admin can set custom price (different from base product)
 - Admin can reject request with reason
 - Customer is notified of approval/rejection via email

3.5 Inventory Management

FR-20: Update Product Stock

- **Requirement:** Administrators shall be able to update product stock levels
- **User Story:** US-16
- **Description:** Business owner must be able to adjust inventory quantities
- **Acceptance Criteria:**
 - Admin can view current stock for each product
 - Admin can edit stock quantity
 - Admin can set stock minimum threshold
 - Change is saved immediately
 - Stock update is logged (who, when, quantity change)
 - Low stock warnings available (optional)

FR-21: Validate Stock Availability

- **Requirement:** The system shall prevent orders of out-of-stock items
- **User Story:** US-08
- **Description:** Customers cannot order products that are out of stock
- **Acceptance Criteria:**
 - Products with stock = 0 appear as "Out of Stock"
 - Out-of-stock items cannot be added to order
 - Error message explains unavailability
 - Customer option to be notified when back in stock (optional)

3.6 Shipping Configuration

FR-22: Configure Shipping Settings

- **Requirement:** Administrators shall configure shipping costs by province
- **User Story:** US-17
- **Description:** Business owner sets up shipping cost structure
- **Acceptance Criteria:**
 - Admin sets base province of business
 - Admin enters list of provinces where they ship
 - Admin sets shipping cost for each province
 - Same province = \$0 additional cost
 - Different province = configured cost
 - Admin can edit costs at any time
 - Changes apply to new orders only (not existing)

3.7 Reporting & Analytics

FR-23: Sales Reports

- **Requirement:** Administrators shall be able to view sales reports
 - **User Story:** US-18
 - **Description:** Business owner needs insights into sales performance
 - **Acceptance Criteria:**
 - Reports show total orders by time period
 - Reports show revenue by period
 - Reports can be filtered by date range
 - Reports show breakdown by product category
 - Reports show top-selling products
 - Reports can be exported (PDF/Excel optional)
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4. Non-Functional Requirements

4.1 Performance Requirements

NFR-1: Page Load Time

- **Requirement:** All pages shall load in less than 3 seconds on standard internet (5 Mbps)
- **Measurement:** Average page load time across all pages
- **Target:** < 3 seconds
- **Method:** Monitor with web analytics tools

NFR-2: Search Performance

- **Requirement:** Product search shall return results within 500 milliseconds
- **Measurement:** Time from search query to results display
- **Target:** < 500ms for 10,000 products
- **Method:** Load testing

NFR-3: Concurrent Users

- **Requirement:** System shall support minimum 1,000 concurrent users
- **Measurement:** Number of simultaneous active users
- **Target:** 1,000+ without degradation
- **Method:** Load testing and stress testing

NFR-4: Database Response Time

- **Requirement:** Database queries shall respond in less than 100 milliseconds
- **Measurement:** Average query response time
- **Target:** < 100ms for standard queries
- **Method:** Database monitoring

4.2 Scalability Requirements

NFR-5: Horizontal Scaling

- **Requirement:** Application shall be designed to scale horizontally
- **Description:** Additional servers can be added to handle increased load
- **Implementation:** Stateless application design, load balancing

NFR-6: Database Scaling

- **Requirement:** Database shall support growth to 1 million records
- **Description:** Proper indexing and optimization for large datasets
- **Implementation:** Database optimization, read replicas (if needed)

4.3 Availability Requirements

NFR-7: System Uptime

- **Requirement:** System shall achieve 99% uptime (excluding planned maintenance)
- **Measurement:** (Total time - Downtime) / Total time
- **Target:** 99% uptime annually (≈ 3.65 days downtime per year max)
- **Method:** Monitoring tools, redundancy

NFR-8: Backup & Recovery

- **Requirement:** Data backups shall occur daily with recovery time objective of 1 hour
- **Measurement:** Latest backup available and recovery time
- **Target:** Daily backups, < 1 hour recovery time
- **Implementation:** Automated backup scripts, tested recovery procedures

4.4 Security Requirements

NFR-9: Password Security

- **Requirement:** Passwords shall be encrypted using bcrypt or equivalent
- **Salt Rounds:** Minimum 10 rounds
- **No plaintext storage:** All passwords hashed before storage

NFR-10: HTTPS/SSL

- **Requirement:** All data transmission shall be encrypted using HTTPS
- **Certificate:** Valid SSL/TLS certificate required
- **Enforcement:** Redirect HTTP to HTTPS

NFR-11: Data Protection

- **Requirement:** Sensitive data shall be protected from unauthorized access
- **Measures:**
 - Database encryption at rest
 - Role-based access control (RBAC)
 - Audit logging of admin actions
 - Regular security updates

NFR-12: Session Management

- **Requirement:** User sessions shall timeout after inactivity
- **Timeout Duration:** 30 minutes of inactivity
- **Implementation:** Server-side session management, automatic logout

NFR-13: Input Validation

- **Requirement:** All user input shall be validated and sanitized
- **Protection Against:** SQL injection, XSS, CSRF attacks
- **Implementation:** Input validation library, parameterized queries

4.5 Usability Requirements

NFR-14: Mobile Responsiveness

- **Requirement:** Application shall be responsive on all screen sizes
- **Breakpoints:**
 - Mobile: 320px - 767px
 - Tablet: 768px - 1023px
 - Desktop: 1024px and above
- **Implementation:** Mobile-first design approach

NFR-15: Accessibility

- **Requirement:** Application shall meet WCAG 2.1 AA accessibility standards
- **Features:**
 - Keyboard navigation support
 - Alt text for all images
 - Proper color contrast (4.5:1 for text)
 - Screen reader compatibility

NFR-16: User Interface Consistency

- **Requirement:** User interface shall be consistent across all pages
- **Implementation:**
 - Unified color scheme
 - Consistent button styling
 - Standard navigation menus
 - Consistent error messages

4.6 Compatibility Requirements

NFR-17: Browser Compatibility

- **Requirement:** Application shall work on modern browsers
- **Supported:**
 - Chrome (latest 2 versions)
 - Firefox (latest 2 versions)
 - Safari (latest 2 versions)
 - Edge (latest 2 versions)
- **Testing:** Cross-browser testing required

NFR-18: Device Compatibility

- **Requirement:** Application shall work on desktop, tablet, and mobile devices
- **Devices:**
 - iPhone 8+
 - Android devices (Android 8+)
 - iPad (5th generation+)
 - Desktop computers

4.7 Maintainability Requirements

NFR-19: Code Quality

- **Requirement:** Code shall follow best practices and style guidelines
- **Standards:**
 - Code comments for complex logic
 - DRY (Don't Repeat Yourself) principle
 - SOLID principles
 - Maximum 20 lines per function (where possible)

NFR-20: Documentation

- **Requirement:** System shall be fully documented
- **Documentation Types:**
 - Code comments
 - API documentation
 - Database schema documentation

- User manual
- Admin manual
- Deployment guide

NFR-21: Testing Coverage

- **Requirement:** Code shall have minimum 80% test coverage
 - **Test Types:**
 - Unit tests (individual functions)
 - Integration tests (component interactions)
 - End-to-end tests (user workflows)
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5. Data Requirements

5.1 Data Entities

Users

- **Data to Store:** id, email, password_hash, name, phone, address, province, registration_date, status
- **Relationships:** Has-Many orders, Has-One profile
- **Retention:** Keep indefinitely

Products

- **Data to Store:** id, name, description, price, category, stock, photos, allows_customization, status, creation_date
- **Relationships:** Belongs-To category, Has-Many order_details
- **Retention:** Keep indefinitely (soft delete)

Orders

- **Data to Store:** id, customer_id, reference_number, date, items, total, shipping_cost, status, delivery_address, payment_status
- **Relationships:** Belongs-To customer, Has-Many order_details
- **Retention:** Keep for minimum 3 years (business requirement)

Order Details

- **Data to Store:** id, order_id, product_id, quantity, price_unit, customization_specs

- **Relationships:** Belongs-To order and product
- **Retention:** Keep indefinitely (linked to orders)

Customizations

- **Data to Store:** id, order_id, product_id, type, details, price_additional, status, approval_date
- **Relationships:** Belongs-To order and product
- **Retention:** Keep indefinitely

5.2 Data Validation Rules

Field	Validation Rule
Email	Must be valid email format, must be unique
Password	Minimum 8 characters, must contain uppercase, lowercase, number
Phone	Must be valid phone format (10-15 digits)
Price	Must be decimal, > 0
Stock	Must be integer, >= 0
Province	Must be from predefined list
Product Name	Required, max 100 characters
Quantity	Must be integer, >= 1

5.3 Data Privacy & Retention

Data Collection

- Only collect data necessary for business operations
- Inform users what data is collected and why

Data Storage

- Passwords hashed immediately upon receipt
- Personal information stored securely
- No sensitive data in logs

Data Retention

- Customer data: 3 years after last activity
- Order data: 3 years (legal requirement)
- Backup data: 30 days retention

Data Deletion

- Customer can request account deletion
 - Process: Anonymize personal data, keep order records for legal compliance
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6. User Interface Requirements

6.1 Layout & Navigation

Header/Navigation

- **Logo:** Brand logo on left, clickable to home
- **Search Bar:** Centered search for products
- **Navigation Links:** Home, Categories, Account, Cart
- **Mobile:** Hamburger menu for small screens

Footer

- **Content:** Contact info, company info, links, copyright
- **Links:** About, Contact, Privacy Policy, Terms of Service
- **Social:** Social media links (optional)

6.2 Customer Interface Pages

Home Page

- **Featured Products:** Display 4-6 featured products
- **Categories:** Show 3 category cards with images
- **Recent Orders:** Show recent orders by other customers (anonymous)
- **Call-to-Action:** Clear button to start shopping

Product Catalog Page

- **Filters Sidebar:** Category, price range, availability
- **Product Grid:** 3-4 products per row (desktop), 1-2 (mobile)

- **Product Card:** Image, name, price, star rating, "Add to Cart" button
- **Pagination:** Show 20 products per page

Product Detail Page

- **Main Image:** Large product photo
- **Thumbnails:** Gallery navigation
- **Information:** Name, price, stock status, description
- **Customization:** Customization option if available
- **Add to Cart:** Quantity selector and add button
- **Related Products:** Show 4 similar products

Shopping Cart Page

- **Cart Items:** List of added products with images
- **Quantity:** Editable quantity for each item
- **Subtotal:** Calculated subtotal per item
- **Totals:** Subtotal, shipping, total
- **Checkout Button:** Prominent checkout button

Checkout Page

- **Order Summary:** Review items and total
- **Contact Information:** Name, email, phone, address, province
- **Shipping Cost:** Calculated and displayed
- **Order Button:** Final confirmation button
- **Confirmation:** Display order reference number

Account Page

- **Profile Info:** Display and edit user information
- **Order History:** List of all orders with status
- **Logout:** Button to log out

6.3 Admin Interface Pages

Dashboard

- **Summary Cards:** Total orders, total revenue, pending customizations
- **Charts:** Sales by category, orders by status

- **Quick Actions:** Add product, view pending orders

Products Management

- **Product List:** Table with product names, stock, status
- **Actions:** Edit, Delete, Add new buttons
- **Search/Filter:** Find products quickly
- **Add Product Form:** Form for creating new products

Orders Management

- **Orders List:** Table with order date, customer, status, total
- **Filters:** By status, date range, customer
- **Order Details:** Modal/page showing full order information
- **Status Update:** Dropdown to change status
- **Customer Info:** Contact details for communication

Shipping Configuration

- **Base Province:** Setting for business location
 - **Province Costs:** Table with provinces and shipping costs
 - **Add/Edit:** Ability to add or modify shipping costs
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7. Security Requirements

7.1 Authentication

- All users must log in to access their data
- Session timeout after 30 minutes of inactivity
- Secure password reset via email
- Failed login attempt logging

7.2 Authorization

- Customers can only view their own orders
- Customers cannot access admin features
- Admin can view all orders and products
- Role-based access control (RBAC)

- HTTPS encryption for all data in transit
- Database encryption at rest
- Regular security audits
- No sensitive data in error messages
- Audit logs for admin actions

7.4 Compliance

- GDPR compliance (if applicable)
 - Data protection regulations
 - Terms of Service & Privacy Policy required
 - Secure backup procedures
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8. Business Rules Enforcement

All 10 business rules must be enforced by the system:

BR	Rule	Implementation
BR-01	Products in 3 categories	Dropdown with 3 options when creating product
BR-02	Only Active products shown	Filter in queries: WHERE status = 'Active'
BR-03	Product requires fields	Mandatory form fields with validation
BR-04	No orders for stock = 0	Validation: IF stock > 0 THEN allow order
BR-05	Order needs min 1 item	Validation: IF item_count >= 1 THEN allow submit
BR-06	Unique reference number	Generate with timestamp + random string
BR-07	Extra cost if different province	Calculate: IF customer_province ≠ business_province THEN add_shipping_cost
BR-08	Customization needs approval	Status workflow: SOLICITADO → APROBADO
BR-09	Mark paid to start production	Status change: CONFIRMADO → PAGADO → EN_PROCESO
BR-10	Only admin can create/edit	RBAC: Only admin_role can access these endpoints

9. Acceptance Criteria Summary

All requirements are considered complete when:

- ☐ All functional requirements are implemented
- ☐ All non-functional requirements are met (performance, security, etc.)
- ☐ All 18 user stories are tested and approved
- ☐ All 10 business rules are enforced
- ☐ All acceptance criteria for each requirement are met
- ☐ Testing coverage >= 80%

- ☐ No critical or high-priority bugs remain
 - ☐ Documentation is complete
 - ☐ User acceptance testing is passed
 - ☐ Stakeholder sign-off obtained
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10. Change Control

Any changes to these requirements must be:

1. Documented in a Change Request
 2. Reviewed by Project Manager and Business Owner
 3. Impact assessed (scope, timeline, cost)
 4. Approved before implementation
 5. Tracked in version control
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11. Sign-Off

Approval by Stakeholders

Role	Name	Signature	Date
Business Owner	_____	_____	_____
Project Manager	_____	_____	_____
Technical Lead	_____	_____	_____
QA Lead	_____	_____	_____

Document Control:

- Version 1.0 - Initial draft
- Last Updated: June 2026
- Next Review Date: August 2026

Appendices:

- A. System Architecture Diagram

- B. Data Flow Diagram
- C. Use Case Diagram
- D. Class Diagram
- E. Database Schema

End of Document